UNIT 4 - Previous Year Questions Solved

Question: Discuss in details the need and importance of Food processing operations

Answer:

1. Food preservation (prevents spoilage)
    - Food preservation refers to the different techniques applied to food to prevent it from spoiling.
    - It involves destruction of microorganisms, reducing/eliminating moisture content, and altering temperature/conditions that favor microbial growth to retard growth and replication.
2. Food safety and shelf life improvement
    - Sterilization destroys all pathogenic and spoilage microorganisms, inactivates enzymes by heating, and gives shelf life of more than two years.
    - Pasteurisation heats food (usually liquid) to or below boiling point for a defined time to destroy pathogens, reduce bacteria, inactivate enzymes, and extend shelf life.
3. Controlling microbial growth by water control (water activity concept)
    - Shelf life depends on availability of water for microbial, enzymatic, or chemical activity, measured as water activity (aw) from 0 to 1.
    - Spoilage microorganisms are generally inhibited below aw 0.6; if pH is less than 4.6, microorganisms are inhibited below aw 0.85.
4. Room temperature stability (Intermediate Moisture Foods)
    - Binding water preserves intermediate moisture foods (cookies, cake, bread) by reducing water availability for deteriorative reactions.
    - Water is immobilized using permissible humectants like glycerol, glycols, sugars, salts; IMFs have aw 0.6 to 0.85 and are stable at room temperature.
5. Chemical control for preservation
    - Addition of chemicals inhibits microbial growth; salt, sugars, wood smoke and some spices also inhibit microorganisms.
    - Almost every food (except egg whites and soda crackers) has pH less than 7; foods are categorized by pH as high acid, acid, medium acid, low acid.
6. Maintaining quality during processing and storage
    - Blanching is a slight heat treatment (hot water/steam), mainly used before freezing/canning vegetables to inactivate enzymes and prevent color/texture changes, off-flavors, and nutritional loss during frozen storage.
7. Cold preservation benefits
    - Refrigeration below 4°C controls microbial growth, reduces respiration rate of fruits/vegetables, retards spoilage reactions, and extends storage life.
    - Freezing (commercial under -18°C) almost stops microbial growth and can give shelf life of one year under ideal conditions.
8. Preservation by moisture removal
    - Evaporation removes water partially from liquid foods by boiling; vacuum evaporation helps retain flavor and reduces weight, storage/transport costs, increases solids, and provides convenience.

    - Dehydration removes water nearly completely from solid foods and is one of the oldest preservation methods (traditionally sun).
9. Improving texture and product properties
    - Homogenization stabilizes emulsions by reducing droplet size through intense shearing forces, improving texture and functional properties.
    - Extrusion compresses food into a semi-solid mass and forces it through a die to create desired texture/shape and provide variety of textured foods.
10. Combination processing (Hurdle technology)
    - Developed to meet demand for natural, fresh-like foods using mild preservation techniques with low-level combinations of hurdles.
    - Common hurdles include pasteurization, water activity, salt, blanching, freezing, pH, preservatives, refrigeration, irradiation, and MAP.

Question: Explain with examples the various methods for food preservations.

Answer:

1. Food preservation (meaning + principle)
    - Food preservation refers to the different techniques that are applied to food to prevent it from spoiling.
    - The science behind food preservation involves:
        - Destruction of microorganisms responsible for causing food spoilage
        - Reducing/eliminating the water (moisture) content from food
        - Altering the temperature and other conditions that favor microbial growth, thereby retarding microbial growth and replication
2. Drying
    - Foods like fruits, vegetables, tubers (cassava, potatoes) can be preserved by drying.
    - Before drying: enough sunshine should be present and foods should be sliced into small pieces to dry faster.
    - Examples:
        - Dried fruits eaten directly (example: dried jackfruit)
        - Dried cassava ground into flour and used later
3. Smoking
    - Smoking meat and fish is a highly recommended method for prolonging storage life.
    - Fish is first cooked over a high fire and then smoke-dried for 1 to 5 days (and nights) over a low fire.
    - Examples (storage life):
        - Fresh-dried fish keeps up to a week
        - Hard-dried fish keeps for several months
4. Salting
    - Salting prolongs shelf life for a few days.
    - Salt takes out moisture and retards microbial growth and replication.
5. Boiling
    - Boiling kills food microbes.
    - Perishable foods can be boiled, cooled, kept in clean containers, and then used within a day.

Question: Classify food packaging methods, and add an explanatory note on the novel food packaging methods

Answer:

1. Classification of food packaging methods
   1.1 Tech enabled solutions

   - Fibre based materials
   - Robotics
   - Smart packaging
   - 3D printing
   - Emotional engagement
   - Vintage inspired designs
   - Transparent and clear labeling
   - Increased portability
   - Personalization
   - Minimal designs
   - Gradient coloring
   - Sustainability

   1.2 Creative food packaging

   - Oxygen scavengers
   - CO_2 generators and scavengers
   - Ethylene scavengers
   - Seltzer packaging
   - Butter
   - Dried fruit

2. Novel food packaging methods (explanatory note: Smart food packaging)

- Retain integrity and actively prevent food spoilage (shelf-life)
- Enhance product attributes like look, taste, flavor, aroma etc.
- Respond actively to changes in product or package environment
- Communicate product information, product history or condition to user
- Assist with opening and indicate seal integrity
- Confirm product authenticity and act to counter theft

  Example (given in PDF)- Film membrane sensor:

  - Checks food freshness without opening
  - Detects bacterial growth through pH
  - Safe in direct contact with food
  - Easily integrated in food packaging

Question: Describe Good Manufacturing Practices applicable to food and the advantages of GAP and GLP.

Answer:

1. Good Manufacturing Practices (GMP) applicable to food
    - Good Manufacturing Practices (GMPs) describe the methods, equipment, facilities, and controls for producing processed food.
    - They are the minimum sanitary and processing requirements for producing safe and wholesome food.
    - They are an important part of regulatory control over the safety of the nation's food supply.

    10 principles of GMP:

    - Create Standard Operating Procedures (SOPs)
    - Enforce / Implement SOPs and work instructions
    - Document procedures and processes
    - Validate the effectiveness of SOPs
    - Design and use working systems
    - Maintain systems, facilities, and equipment
    - Develop job competence of workers
    - Prevent contamination through cleanliness
    - Prioritize quality and integrate into workflow
    - Conduct GMP audits regularly

    Supporting concepts in GMP:

    - The 5M stands for:
        - Machine (equipment)
        - Medium (environment)
        - Method (process)
        - Material (raw materials)
        - Workforce
    - The 5 P's of GMP:
        - Products
        - Processes
        - Procedures
        - Premises
        - People

2. Advantages of GAP (Good Agricultural Practices)

    - GAP is a voluntary certification program which verifies through an audit that sound food safety practices are being used.
    - It reduces the risk of microbial contamination in fruits, vegetables and nuts and aims to make sure these foods are safe for you to eat.
    - It enables producers in making their systems more sustainable through improved system implementation.
    - It provides visibility of the certified producers on the global database giving access to local and global customers, markets, suppliers and retailers.

- It gives buyers a strong sense of trust based on the assurance of having a certified system in place.
- GAP certification opens new market opportunities.
- It reduces the risk of foodborne illnesses and increases consumer trust.
- It shows that on-farm food safety is at the forefront and demonstrates care about produce quality.

3. Advantages of GLP (Good Laboratory Practices)

In the food industry, GLP is a quality management system used to control:

- color additives
- food additives
- food contamination limits
- food packaging

Question: What is thermal food processing? Explain its advantages and disadvantages. (6 marks)

Answer:

1. Thermal food processing (meaning)
    - Sterilization: destroys all pathogenic and spoilage microorganisms in foods and inactivates enzymes by heating.
    - Pasteurisation: process of heating a food (usually a liquid) to or below its boiling point for a defined period of time.
2. Advantages of thermal food processing
    - Destroys pathogens and spoilage microorganisms in foods.
    - Inactivates enzymes by heating.
    - Sterilization enables food to have a shelf life of more than two years.
    - Pasteurisation reduces the number of bacteria and extends the shelf life of a food product.
    - Permanent stability (shelf life of about two years) is obtained with foods that can withstand prolonged heating (example: bottled juices).
3. Disadvantages of thermal food processing
    - Greater loss of flavor occurs when foods are exposed to a longer time-temperature relationship.
    - Some foods get only temporary stability (limited shelf life) because prolonged heating would destroy quality (example: milk).
    - Such foods usually require subsequent refrigeration.

Question: Discuss the importance of food preservation, processing, and packaging in the food industry. (6 marks)

Answer:

1. Importance of food preservation
    - Food preservation refers to the different techniques applied to food to prevent it from spoiling.

    - It involves:
        - Destruction of microorganisms responsible for causing food spoilage
        - Reducing/eliminating water (moisture) content from food
        - Altering temperature and other conditions that favor microbial growth to retard growth and replication
    - Advantages of preservation:
        - Safety: prevents growth of germs and microorganisms and helps prevent food poisoning and foodborne illnesses.
        - Quality: helps maintain nutritional and sensory quality of food.
        - Availability: ensures food is available in remote areas and during the offseason.
        - Sustainability: reduces food waste and saves resources like water, energy, and land used in food production.
2. Importance of food processing
    - Food production involves preparing food where raw materials are converted into ready-made food products.
    - It includes sections like cleaning, packing, segregating, sorting, preparing, adding ingredients in correct proportions, presenting.
3. Importance of food packaging
    - Packaging means wrapping or bottling of products to make them safe from damages during transportation and storage.
    - It keeps a product safe and marketable.
    - It helps in identifying, describing, and promoting the product.
    - Effects of poor packaging (importance of proper packaging):
        - Increases the risk of shrinkage
        - Contents may get damaged in transit
        - Causes contamination through leakage
        - Leads to product spoilage or damage
        - Contributes to product loss through misidentification
        - Loss of contents
        - Becomes too ugly to sell
        - Decreases shelf life and reduces sales

Question: Compare and contrast refrigeration and freezing as methods of food preservation. (4 marks)

Answer:

Method	**Refrigeration**	**Freezing**
Temperature	Below 4°C	Commercial freezers under -18°C (home freezers around -10°C)

Effect on microorganisms	Controls growth of microorganisms in foods	Growth of microorganisms is almost stopped (microbial reactions still occur but over longer time)
Other effects	Reduces respiration rate of fruits and vegetables and retards reactions that promote spoilage	Water in food forms ice crystals; quality depends on slow freezing and rapid freezing; enzymatic reactions still take place during frozen storage; uncooked fruits and vegetables must be blanched before freezing to prevent enzymatic reactions
Uses / Shelf life	Reduce spoilage during distribution of perishable foods; increase holding period between harvesting and processing; extend storage life of commercially processed foods	Under ideal conditions, frozen foods can have a shelf life of one year; exposure to warmer temperatures causes heat shock (ice melts and reforms into larger crystals, example: gritty ice cream texture)

Question: "Understanding food processing principles and operations is key to designing efficient systems that meet safety, nutritional, and economic demands of the modern food industry". Justify the statement (16 marks)

Answer:

1. Food processing operations decide how raw materials become safe, usable products
    - Food processing refers to operations performed on food materials to convert them into a consumable form.
    - It includes cleaning, packing, segregating, sorting, preparing, adding ingredients in correct proportions, and presenting.
2. Preservation is required to prevent spoilage and ensure safety
    - Food preservation refers to techniques applied to food to prevent spoilage.
    - It works by destruction of microorganisms, reducing/eliminating moisture, and altering temperature/conditions to retard microbial growth and replication.
3. Thermal processing supports food safety and shelf life
    - Sterilization destroys all pathogenic and spoilage microorganisms and inactivates enzymes by heating, giving shelf life of more than two years.
    - Pasteurisation heats food (usually liquid) to or below boiling point for a defined time to destroy pathogens, reduce bacteria, inactivate enzymes, and extend shelf life.
    - Limitation: longer time–temperature causes greater flavor loss; some foods (milk) get only temporary stability and require refrigeration.

4. Water activity control is a key principle for shelf life extension
    - Shelf life depends on water availability for microbial, enzymatic, or chemical activity, measured as water activity (aw) from 0 to 1.
    - Spoilage microorganisms are generally inhibited below aw 0.6.
    - If pH is less than 4.6, microorganisms are inhibited below aw 0.85.
5. Intermediate moisture foods show efficient room-temperature stability
    - IMFs (cookies, cake, bread) are preserved by binding water and reducing availability for deteriorative reactions.
    - Humectants used: glycerol, glycols, sugars, salts; IMFs have aw 0.6 to 0.85 and are stable at room temperature.
6. Chemical and pH control strengthen preservation systems
    - Addition of chemicals inhibits microbial growth; salt, sugars, wood smoke and some spices also inhibit microorganisms.
    - Almost every food (except egg whites and soda crackers) has pH less than 7; foods are categorized by pH as high acid, acid, medium acid, low acid.
7. Cold processing protects quality and extends storage life
    - Refrigeration below 4°C controls microbial growth, reduces respiration rate of fruits/vegetables, and retards spoilage reactions.
    - Freezing under -18°C almost stops microbial growth; frozen foods can have shelf life of one year under ideal conditions.
    - Blanching before freezing inactivates enzymes that cause color/texture changes, off-flavors, and nutritional loss during frozen storage.
8. Moisture removal improves economy and convenience
    - Evaporation partially removes water from liquid foods; vacuum evaporation retains flavor.
    - Benefits: reduces weight, storage and transport costs, preserves by increasing solids content, and provides convenient foods.
    - Dehydration nearly completely removes water from solid foods and is an old preservation method (traditionally sun).
9. Packaging is essential for protection, shelf life, and marketability
    - Packaging means wrapping or bottling products to protect from damage during transportation and storage.
    - It keeps a product safe and marketable and helps in identifying, describing, and promoting the product.
    - Poor packaging causes shrinkage, transit damage, contamination through leakage, spoilage, misidentification, content loss, reduced shelf life, and reduced sales.
10. Novel (smart) packaging supports modern food industry demands

- Smart food packaging retains integrity and actively prevents spoilage, enhances product attributes, responds to changes in environment, communicates product information/history/condition, assists opening and seal integrity, confirms authenticity, and counters theft.

11. GMP ensures hygienic, controlled, and reliable production systems

- GMP describes methods, equipment, facilities, and controls for producing processed food.
- It provides minimum sanitary and processing requirements for safe and wholesome food and supports regulatory control.

- Key actions include SOPs, documentation, validation, maintenance, worker competence, preventing contamination, integrating quality, and audits.

Justification:

- Efficient food systems require correct selection and control of preservation principles (thermal processing, water activity, moisture removal, pH/chemical control, refrigeration/freezing) along with packaging and GMP to ensure safety, maintain quality, reduce spoilage, improve shelf life, and reduce economic losses while meeting changing consumer needs.

Question: Explain chemical and physical method of food processing (16 marks)

Answer:

1. Chemical methods of food processing
   1.1 Intermediate Moisture Foods (IMF)
   - IMF preservation is achieved by binding the water present in foods.
   - This reduces availability of water for deteriorative reactions.
   - Water is immobilized by adding permissible humectant additives:
     - glycerol, glycols, sugars, salts
   - IMFs possess water activities ranging from 0.6 to 0.85.
   - This enables foods to be stable at room temperature (growth of most microorganisms inhibited).
   - Examples: cookies, cake, bread
2. 1.2 Water activity (aw)
   - Availability of water determines shelf life of foods for microbial, enzymatic, or chemical activity.
   - Water availability is measured as water activity (aw) on a scale of 0 to 1:
     - 0 indicates no water
     - 1 indicates all water
   - Food spoilage microorganisms are generally inhibited when aw is below 0.6.
   - If pH is less than 4.6, microorganisms are inhibited when aw is below 0.85.
3. 1.3 Addition of chemicals
   - Addition of some chemicals inhibits microbial growth in foods.
   - This includes not only those classified as preservatives.
   - Salt, sugars, wood smoke, and some spices also inhibit growth of microorganisms.
4. 1.4 pH control
   - Almost every food, with the exception of egg whites and soda crackers, has a pH value of less than 7.
   - Foods are broadly categorized based on pH into:
     - high acid
     - acid
     - medium acid
     - low acid
5. Physical methods of food processing
   2.1 Sterilization

    - Destroys all pathogenic and spoilage microorganisms in foods.
    - Inactivates enzymes by heating.
    - Food can have shelf life of more than two years.
    - Canned foods are sterilized in a retort (large pressure cooker).
6. 2.2 Pasteurisation
    - Heating food (usually a liquid) to or below its boiling point for a defined period.
    - Purpose:
        - destroy all pathogens
        - reduce number of bacteria
        - inactivate enzymes
        - extend shelf life
7. 2.3 Blanching
    - Slight heat treatment using hot water or steam.
    - Applied mostly to vegetables before canning or freezing.
    - Used before freezing to inactivate enzymes causing:
        - color changes
        - texture changes
        - off-flavors
        - decrease in nutritional value
8. 2.4 Microwaving
    - Food is heated by subjecting it to microwave energy.
    - Microwave ovens are used in the home and in restaurants.
    - Microwaves have sufficient energy to cause changes in molecular structure, generating heat.
    - Microwave heating occurs when polar molecules (like water) attempt to align with changing electric field.
9. 2.5 Frying
    - Food is heated by immersion in hot oil.
    - Temperature is around 150°C to 200°C.
    - Purpose:
        - destroys microorganisms
        - produces desirable color, flavor, and crispness
10. 2.6 Refrigeration
    - Temperature should be set below 4°C to control growth of microorganisms.
    - Lower temperature reduces respiration rate of fruits and vegetables.
    - Retards reactions that promote spoilage.
    - Used to:
        - reduce spoilage during distribution of perishable foods
        - increase holding period between harvesting and processing
        - extend storage life of commercially processed foods
11. 2.7 Freezing
    - Commercial freezers operate under -18°C (home freezers around -10°C).
    - Growth of microorganisms is almost stopped.
    - Microbial reactions still occur but over longer time.
    - Frozen foods can have shelf life of one year under ideal conditions.
12. 2.8 Irradiation
    - Food is exposed to gamma radiation (from cobalt-60) or electron beam.
    - Ionizing radiation passes through food causing:

        - microbial death
        - enzyme inactivation
    - Can improve shelf life by:
        - killing insects and parasites
        - delaying ripening and sprouting
13. 2.9 Evaporation
    - Partial removal of water from liquid foods by boiling.
    - Under vacuum, boiling avoided and flavor qualities retained.
    - Purpose:
        - reduce weight
        - reduce storage and transport costs
        - preserve by increasing solids content
        - provide convenient foods
14. 2.10 Dehydration
    - Nearly complete removal of water from solid foods.
    - One of the oldest methods of food preservation (traditionally sun).
    - Used for the same reasons as evaporation.
15. 2.11 Emulsion
    - System containing two liquid phases that do not mix.
    - One phase distributed throughout the other as very small droplets.
    - Types:
        - oil in water (O/W)
        - water in oil (W/O)
16. 2.12 Homogenization
    - Used to stabilize an emulsion.
    - Droplet size reduced and number of droplets increased by intense shearing forces.
    - Applied to:
        - change functional properties
        - improve texture of emulsions
17. 2.13 Extrusion
    - Food is compressed and worked to form a semi-solid mass.
    - Forced through a die to create desired texture or shape.
    - Used to provide greater variety of textured foods to consumers.

Question: Describe three levels of packaging the food products (10 marks)

Answer:

1. Primary packaging
    - Packaging means the wrapping or bottling of products to make them safe from damages during transportation and storage.
    - Primary packaging is the packaging that contains the product.
    - It is all of the packaging which surrounds the product when the consumer takes it home.
    - Example: for a multipack of crisps
        - the individual bags are primary packaging

        - the large bag into which the separate packs are packed is also primary packaging
2. Secondary packaging
    - Secondary packaging includes boxes or containers containing specific quantities of primary packages.
    - It is used to group packs together for ease of handling.
    - Example: for the crisps multipacks
        - several multipacks are packed into printed corrugated cases
        - the corrugated case is the secondary packaging
    - Other examples of secondary packaging:
        - shrinkwrap film
        - corrugated board trays
        - thermoformed plastic trays used for shelf-ready packaging
3. Tertiary packaging
    - Tertiary packaging includes pallets and large shipping containers for storing and warehousing.
    - It is used to collate secondary packs for ease of transport.
    - One of the most common forms of tertiary packaging is the pallet.
    - Pallet is used along with:
        - stretch wrap film
        - a label (to secure the secondary packs to the pallet and provide a ready means of identification)
    - Other examples of tertiary packaging:
        - roll cages
        - crates
4. Packaging factors considered (supports packaging level decisions)
    - 1960s: convenience
    - 1970s: light weight
    - 1980s: source reduction
    - From 1990s: environmental impact

Question: Explain the effects of poor packaging on food products, add a note on smart packing (10 marks)

Answer:

1. Effects of poor packaging on food products
    - Increases the risk of shrinkage.
    - Contents may get damaged in transit.
    - Leads to contamination through leakage.
    - May cause product spoilage or damage.
    - Contributes to product loss through misidentification.
    - Causes loss of contents.
    - Makes the product too ugly to sell.
    - Decreases shelf life and reduces sales.
2. Smart packing (Smart food packaging) – note
    - Retains integrity and actively prevents food spoilage (shelf-life).
    - Enhances product attributes like look, taste, flavor, aroma etc.

    - Responds actively to changes in product or package environment.
    - Communicates product information, product history or condition to user.
    - Assists with opening and indicates seal integrity.
    - Confirms product authenticity and acts to counter theft.
3. Example (given)
    - Film membrane sensor:
        - Checks food freshness without opening
        - Detects bacterial growth through pH
        - Safe in direct contact with food
        - Easily integrated in food packaging